

PAPER_705: NGC 3603 Star Cluster (Variant 2): Bok Globule Secondary SF UQFF

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Abstract

NGC 3603 is the most massive young (< 1 Myr) star cluster known in the Milky Way, located in the Carina arm at 20,000 ly. With $M_{cluster} \approx 4 \times 10^5 M_{\odot}$, it drives cavities in surrounding molecular gas via OB stellar winds. This variant-2 analysis emphasizes secondary Bok-globule-triggered star formation and wind-cavity expansion dynamics under the UQFF MUGE framework.

1. System Parameters

Parameter	Value	Description
M_{init}	$4 \times 10^5 M_{\odot}$	Initial cluster mass
r_{clust}	8.998×10^{15} m	Cluster radius
$\dot{M}_0$	0.10	Bok globule SF fraction
τ_{SF}	3.156×10^{13} s	SF timescale (1 Myr)
P_0	0.10	Wind pressure suppression

2. Master UQFF Gravity Equation

$$g_{v2}(t) = \frac{GM(t)}{r^2} (1 + H_0 t) (1 - P(t)) (1 + f_{TRZ}) + q(v \times B) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}} \right) \times 10^{-12}$$

2.1 Dynamic Mass (Secondary SF)

$$M(t) = M_{init} (1 + \dot{M}_0 e^{-t/\tau_{SF}})$$

2.2 Wind Cavity Pressure Reduction

$$P(t) = P_0 e^{-t/\tau_{exp}}; \quad (1 - P) \approx 0.939 \text{ at } t = 0.5 \text{ Myr}$$

3. Numerical Result

$$g_{NGC3603,v2}(t = 0.5 \text{ Myr}) \approx 1.053 \times 10^{-3} \text{ m/s}^2$$

References

- Crowther et al. (2010), MNRAS, 408, 731
- Harayama et al. (2008), ApJ, 675, 1319
- UQFF Framework, Star-Magic Session 175

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